MRDVS®



S01 Small-Array dToF LiDAR
User Manual

Manual version: V1.0

Release date: 2026-10-21

Revision History

Document Version	Release Date	Description
V1.0	2026-10-21	Initial release

Glossary

Term	Definition
Depth	A depth image in which each pixel represents the distance from the camera to the corresponding point in the scene.
FOV	Field of view: the angular extent of the camera's imaging area, including horizontal (HFOV), vertical (VFOV), and diagonal (DFOV) fields of view.
IR	An infrared amplitude image showing the strength of the received infrared return signal. Pixel values indicate the relative signal strength.
SoC	System on a chip: an integrated circuit used for image processing, depth calculation, and application algorithm execution.
dToF	Direct time of flight: calculates distance by measuring the time taken for a light pulse to travel from the emitter to the target and back to the receiver.
iToF	Indirect time of flight: calculates distance from the phase difference between the emitted modulated light and the received return signal.
IMU	Inertial measurement unit: measures motion information such as angular velocity and linear acceleration.
SLAM	Simultaneous localization and mapping: estimates device position and builds an environmental map in real time in an unknown environment.

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1 Product Overview

1.1 Introduction

The MRDVS S01 is a compact, small-array dToF LiDAR sensor. Its compact size, cost effectiveness, and reliable adaptability to different operating conditions make it suitable for presence detection, position confirmation, near-field obstacle avoidance, blind-spot coverage, and height positioning. It provides a practical and reliable 3D perception solution for a wide range of intelligent mobile platforms.

1.2 Key Features

1.2.1 Accurate Near-Field 3D Perception

Using dToF technology, the S01 captures target depth information over a range of 0.15-2 m. It is suitable for near-field obstacle detection and blind-spot coverage on mobile robots, autonomous forklifts, cleaning robots, and similar equipment.

1.2.2 Compact Design for Flexible Integration

The compact structure fits beneath or beside a robot and in other locations with limited space, supporting the near-field sensing requirements of different types of equipment.

1.2.3 3D Sensing for Near-Field Obstacle Detection

Unlike a conventional single-point distance sensor, the S01 captures depth information at multiple points within its field of view. This enables it to detect the spatial distribution of nearby obstacles for applications such as edge-following and near-field collision avoidance.

1.2.4 Low Power Consumption for Embedded Applications

The low-power design reduces energy consumption and heat dissipation requirements, making it suitable for robots and automation equipment with limited installation space and power budgets.

2 Technical Specifications

2.1 General Specifications

Parameter	S01 Small-Array dToF LiDAR
Operating Principle	dToF
Output Formats	Depth image, intensity image, and 3D point cloud
ToF Resolution / Frame Rate	40 × 30 at 10 fps; maximum 20 fps
ToF Field of View	58° × 45° (±2°)
Distance Range	0.15-2 m at 10%-90% reflectivity
Ranging Accuracy	Over 80% of the image area at 100 klux: 1-sigma error < 2 cm + 1‰ × d
Average Operating Power	< 1 W
Peak Power	Approximately 4 W
Average Standby Power	< 1 W
Laser Wavelength	940 nm
Dimensions	41 × 27 × 21 mm[1]
Weight	Approximately 25 g
Power Supply	Standard version: 5 V Industrial version: 12-24 V
Communication Interface	USB[2]
Ingress Protection	IP54
Operating Temperature	-20°C to 50°C
Storage Temperature	-20°C to 60°C
Operating Humidity	0%-95% RH, non-condensing
Ambient Light Resistance	100 klux [3]
Software Environment	C/C++/ROS SDK
Supported Systems	Windows 7+, Linux, Arm Linux/ROS

Note [1]: Dimensions are for the standard version only. The industrial version has slightly different dimensions. Contact MRDVS FAE for details.

Note [2]: CAN and I/O interfaces can be supported through expansion. Contact MRDVS FAE for details.

Note [3]: Accuracy may be affected at 100 klux. Contact MRDVS FAE for details.

2.2 Interface Definition

Design and select the cable according to the pin assignments and the requirements of the complete system.

2.2.1 Standard Version

The standard S01 uses a 6-pin GH1.25 receptacle, SM06B-GHS-TB(LF)(SN), as its only external interface.



Figure 1. JST connector

The pins are numbered from left to right as shown below.

Pin No.	Pin Name	Direction	Signal Type	Description
1	VCC	-	POWER	Power input: 5 V [1]
2	USB DN	Bidirectional	USB	USB differential data negative (D-)
3	USB DP	Bidirectional	USB	USB differential data positive (D+)
4	GND	-	GND	Ground
5	RSV	-	-	Reserved
6	RSV	-	-	Reserved

Note [1]: A supply voltage of at least 4.8 V and a current capacity of at least 1 A are recommended.

2.2.2 Industrial Version

The industrial S01 uses the same connector, with different pin assignments as shown below.

Pin No.	Pin Name	Direction	Signal Type	Description
1	VCC	-	POWER	Power input: 10-26 V
2	USB DN	Bidirectional	USB	USB differential data negative (D-)
3	USB DP	Bidirectional	USB	USB differential data positive (D+)
4	GND	-	GND	Ground
5	GPIO	Output	GPIO	Output voltage follows the supply voltage
6	RSV	-	-	Reserved

3 Product Information

3.1 Mechanical Drawing (Standard Version)

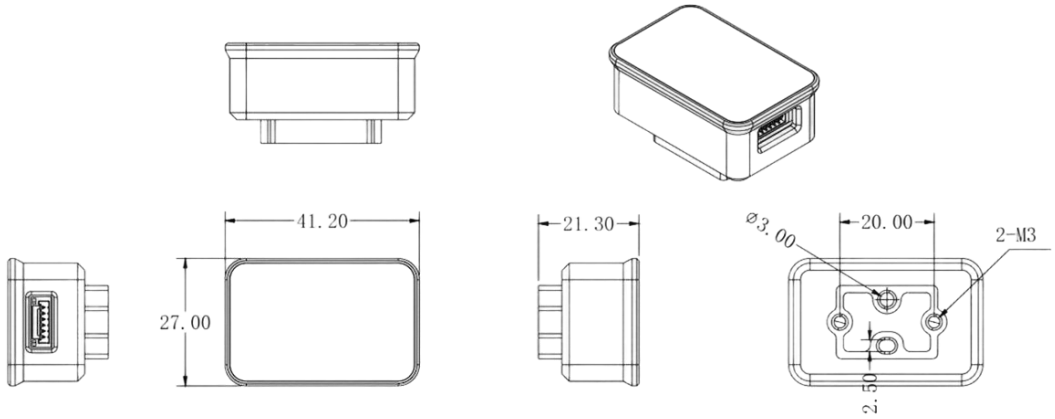


Figure 2. S01 dimensional drawing

The industrial version has slightly different dimensions. Refer to the actual product.

3.2 Camera Components



Figure 3. S01 camera components

4 Camera Integration Guide

4.1 Detection Area

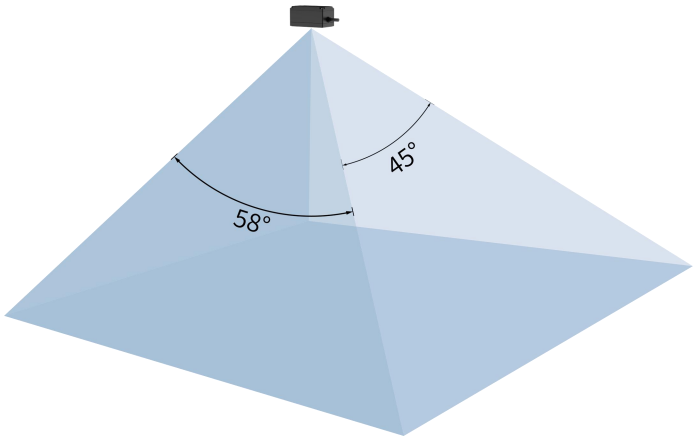


Figure 4. S01 field of view in 3D

4.2 Installation Precautions (Important)

4.2.1 Recommended Installation

Keep the illustrated clearance zone (pink area) completely free of objects when installing the 3D vision camera. Objects entering this zone can interfere with point-cloud data and cause image noise, false obstacle-triggered stops, and abnormal data.

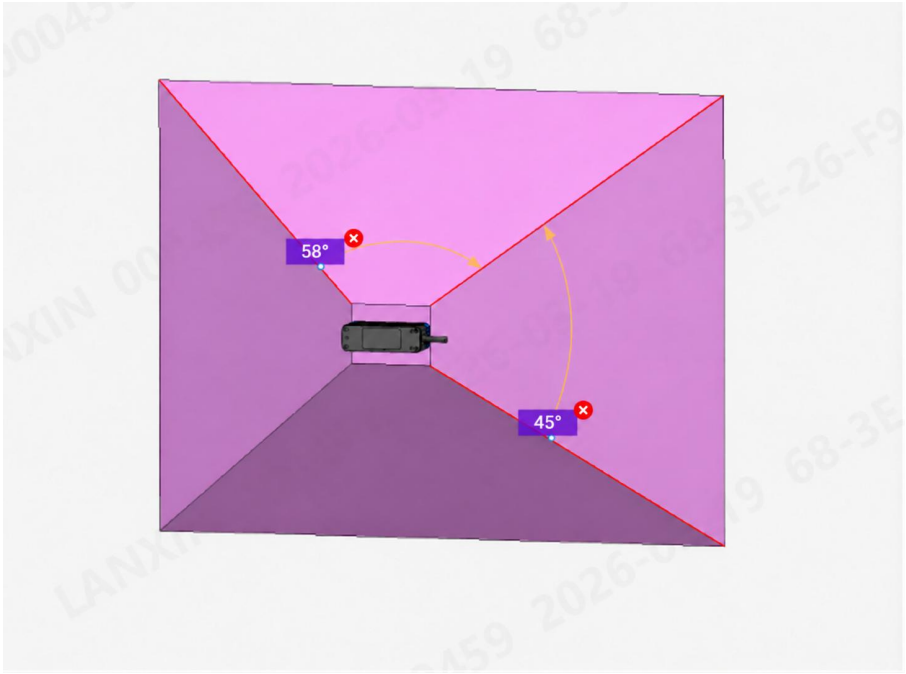


Figure 5. Camera clearance zone in 3D

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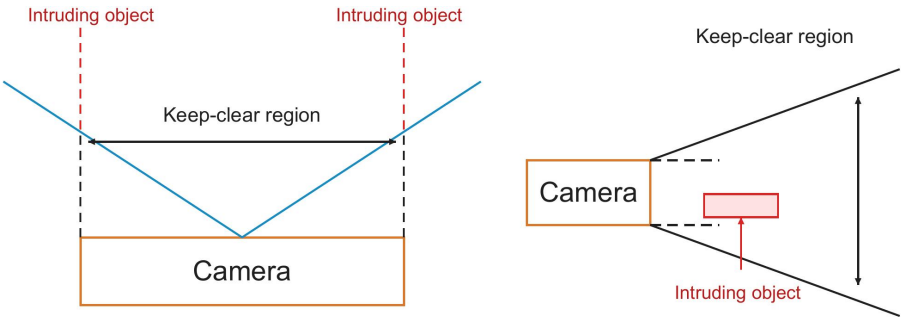


Figure 6. Camera clearance zone in 2D

4.2.2 Incorrect Installation

The examples below show incorrect installations in which objects intrude into the camera clearance zone. Such installations violate the camera's installation requirements and can interfere with point-cloud data, leading to excessive noise, false alarms, and unintended obstacle-triggered stops. Avoid these conditions during installation.

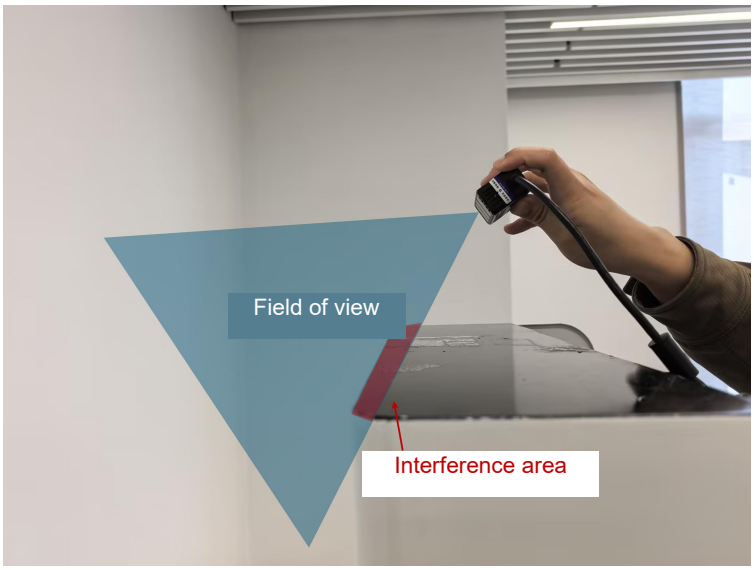


Figure 7. Incorrect installation example 1

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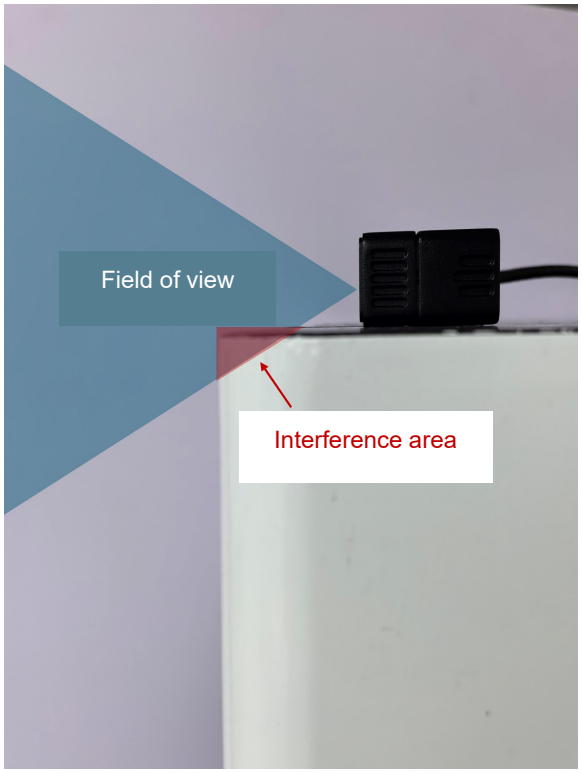


Figure 8. Incorrect installation example 2

4.2.3 Causes of Abnormal Camera Operation

An unsuitable installation environment can interfere with measurement principles such as time of flight and triangulation, causing abnormal operation. Possible causes are described below.

1. Incorrect mechanical installation: obstruction and FOV clipping

Structural frames or ribs may enter the sensor's specified field of view. Misalignment or an opening that is too small can also clip the field of view and cause the sensor to detect the structure as a target. Typical symptoms include fixed blind spots, false short-distance readings, and a reduced field of view. These conditions can cause missed obstacles and incorrect path decisions, reducing perception reliability.

2. Optical interference: near-field infrared crosstalk and reflections

This is the most common cause of failure. A confined installation space can produce abnormal infrared reflections and crosstalk, mainly in two forms:

Internal reflections and multipath interference: infrared light reflects multiple times from the inner surfaces of structural parts before reaching the receiver. This changes the measured travel time and distorts the calculated distance.

Local failure caused by highly reflective surfaces: smooth, shiny materials such as polished metal or mirrors can produce specular reflections. The local return signal may be lost or may saturate the receiver.

3. Optical interference: protective windows

The material and cleanliness of protective glass or another optical window in front of the sensor can cause interference in three main ways:

Ghosting caused by multiple reflections: reflections from the inner and outer surfaces of the window mix with the useful signal, producing double images in the depth map, flying points, or duplicated LiDAR point clouds.

Reduced signal-to-noise ratio caused by contamination or fogging: condensation and dust can accumulate on the window in an enclosed space, preventing the sensor from identifying the useful signal.

Signal attenuation caused by unsuitable materials: the signal may not pass through ordinary glass or acrylic, resulting in a loss of data across the entire image. Window interference can increase measurement errors, damage the lens, and increase maintenance costs. Glass with high optical transmittance may improve performance.

4. Other hidden factors

Mounting stress: an excessively tight clip may slightly deform the lens or shift the optical axis, causing distance errors in a ToF camera.

Thermal effects: poor heat dissipation in an enclosed space can raise the sensor temperature and deform the window, causing ranging drift and, over time, widespread failure.

4.3 SDK

Refer to the MRDVS host application user manual and SDK development guide. The GitHub repository is available at:

https://github.com/Lanxin-MRDVS/CameraSDK

4.4 Operating Environment and Reliability

Item		Specification
Operating Environment	**Temperature**	-20°C to 50°C
	Illuminance	0-100 klux [1]
Storage Environment	**Temperature**	-20°C to 60°C
	Humidity	Relative humidity: maximum 95% RH
Temperature Rise During Normal Operation		Housing temperature rise < 25°C
ESD Rating		Contact discharge: ±4 kV; air discharge: ±8 kV
Radiated Emissions (RE)		Use a ferrite core during operation to meet GB 9254 Class A requirements.
Environmental Compliance		RoHS

Note [1]: Accuracy may be affected at 100 klux. Contact MRDVS FAE for details.

5 Quick Start Guide

5.1 Package Contents

No.	Description	Image
1	S01 camera	

5.2 Connection Test

5.2.1 Power and Configuration

- Connect the camera according to the pin assignments in Section 2.2, Interface Definition.

5.2.2 Run the Software

- Run the supplied host application and acquire images from the camera. Check that the images are clear, without corruption or stuttering, to confirm that the connection is working correctly.
- Refer to Section 5.3, User Interface, or the LxCameraViewer user manual for operating instructions.
- For further assistance, contact MRDVS sales or technical support.

5.3 User Interface

LxCameraViewer supports opening and viewing multiple devices. It can display depth images, intensity images, point clouds, and RGB images, and can read and configure basic camera functions and algorithm parameters. The software also allows users to view and save camera image data, camera settings, and detailed algorithm data.

The initial window has three main areas: the list area (1, blue), menu area (2, green), and image display area (4, red). Double-click the menu area to maximize or restore the window. To move the window, hold the left mouse button in the menu area and drag.

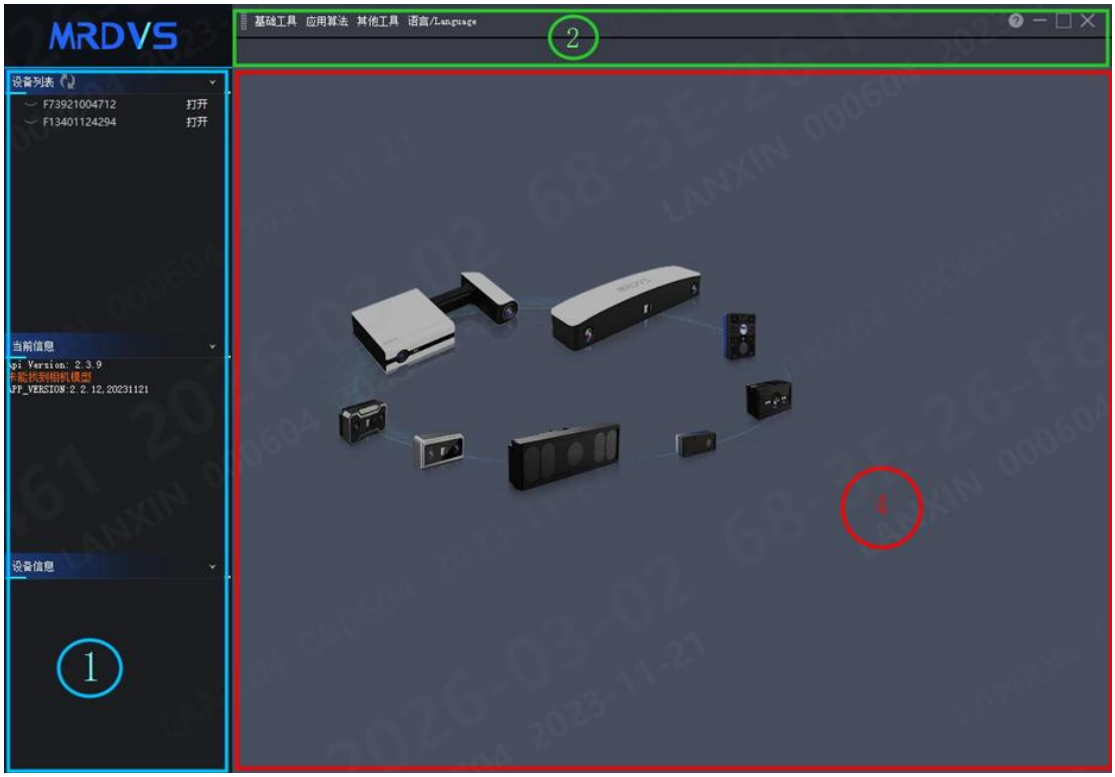


Figure 9. Software interface before opening a camera

After a camera is opened, a function panel appears on the right (area 3). Select Basic Tools, Application Algorithms, or Other from the top menu to access the required functions. Basic Tools includes image display, 2D settings, 3D settings, and filtering.

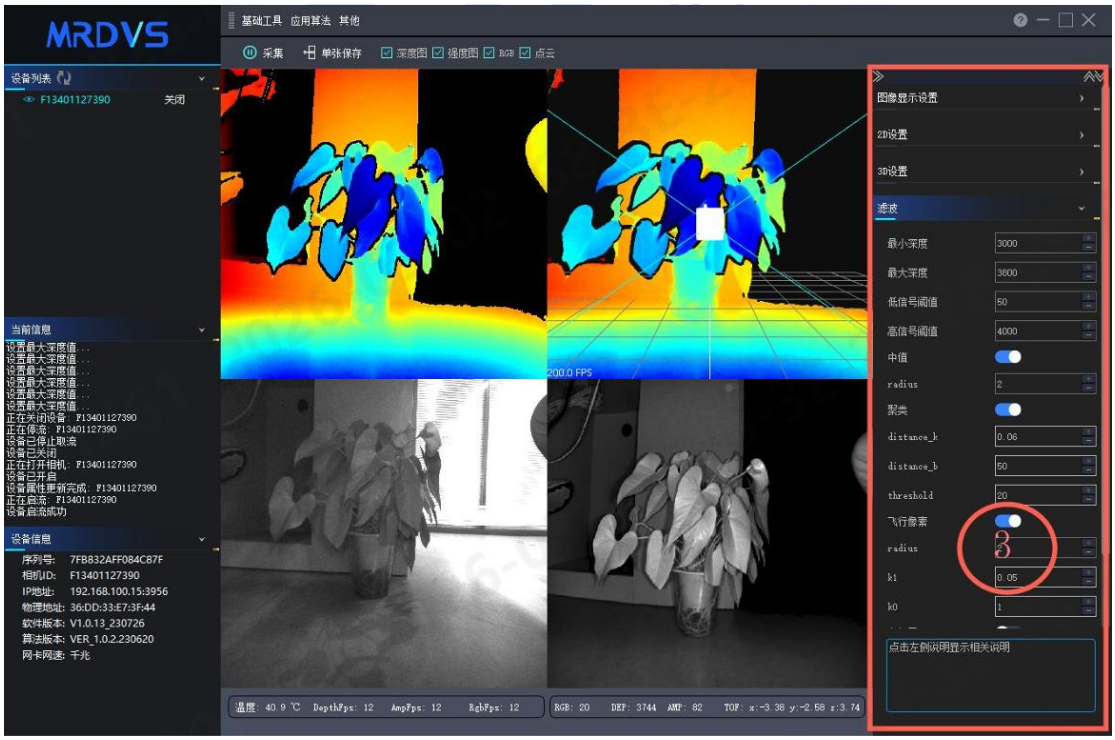


Figure 10. Software interface with a camera open

The device list is in the upper-left corner. Click Open or Close to open or close a camera. The log area in the middle-left displays camera information, software operations, operation results, and streaming results. The device information area in the lower-left displays information about the device currently selected in the device list.

Click Acquire to start or stop image acquisition. The button is disabled until the operation completes. Click Save Frame to save image data from the enabled streams. Select the Depth, Intensity, RGB, and Point Cloud checkboxes to enable the required image formats.

For more details, refer to the LxCameraViewer user manual.

6 Troubleshooting

No.	Issue	Description / Action
1	No data after opening the camera in the host application	Check that the firewall is disabled. Try another computer.
2	Software installation location	Permission settings may apply when the software is installed on the default C: drive.
3	No image, or an unusually small depth image	Update the host application to a version later than Lanxin-MRDVS-1.3.66.0606.
4	Failed to obtain exclusive application access	Ensure that only one user is connected to the camera at a time. Versions 2.4.50.0720 and later use multi-user access control.
5	Connection fails when reopening the camera after changing its IP address	After a failed connection, wait approximately 15 seconds and reconnect.

7 Precautions



Laser Safety

This product emits invisible laser radiation during operation. Avoid exposing eyes to the beam.

The emitted laser meets Class 1 safety requirements. In accordance with EN 60825, it is not hazardous to the human body during normal use.

Figure 11. Laser safety classification

Further Support

http://www.mrdvs.cn/

https://space.bilibili.com/3546755102672963

https://github.com/Lanxin-MRDVS/CameraSDK

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